Lab Report

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Team: E1

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Goal

Print the 3D modelled holders, assemble them with their corresponding components, check for a proper fit, and make any final adjustments.

Discussion

- IR-Sensors Holders

Modification:The original design called for placing a square ring over the sensor, as shown in Figure 1. Dr.Gido provided feedback so we could save printing time. The IR-Sensors Holders were modified, as shown in Figure 2.

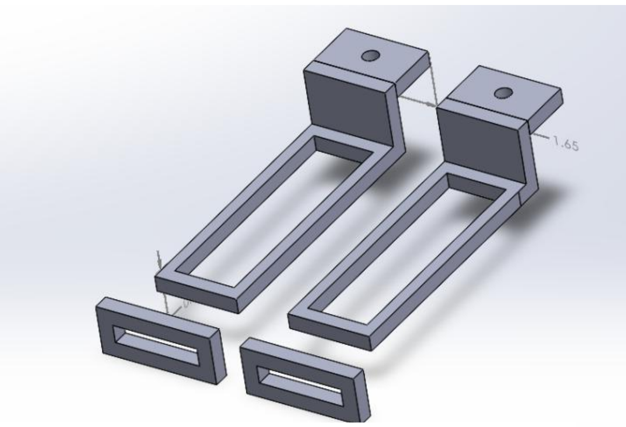

Figure 1 original design of IR-Sensors Holder

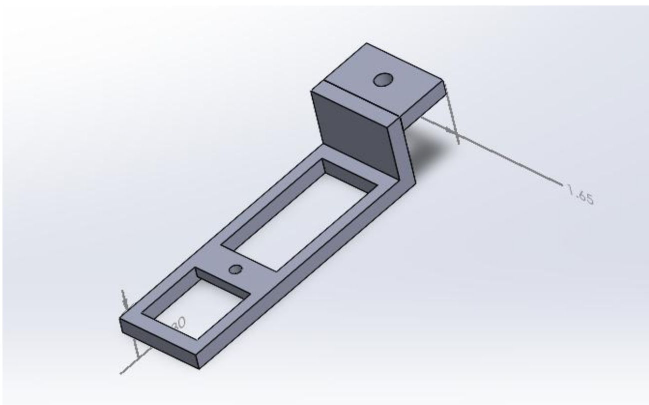

Figure 2 latest design of IR-Sensors Holder

- Ultrasonic-Sensors Holders

Problem: The structure is too fragile to be used, as shown in Figure 3.

Solution: A redesign is required, as shown in Figure 4.

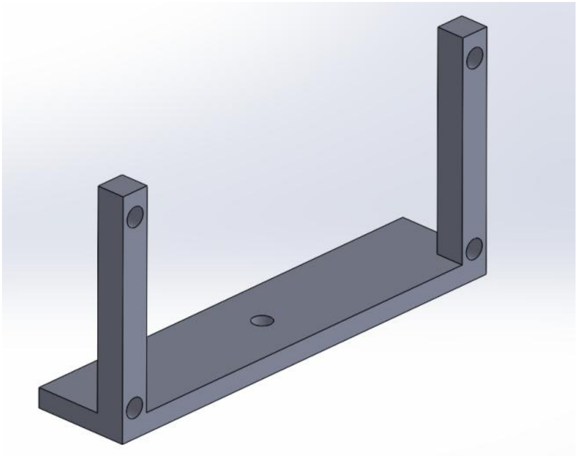

Figure 3 original design of Ultrasonic-Sensors Holder

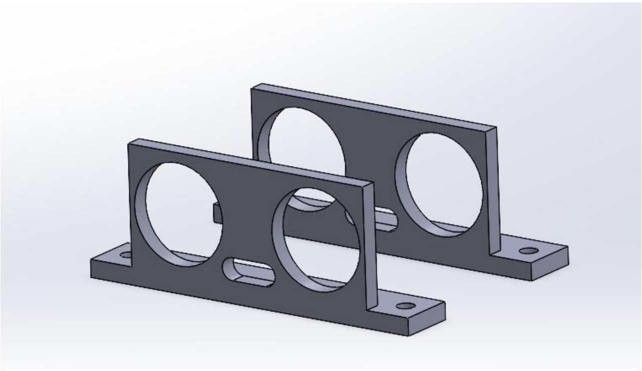

Figure 4 latest design of Ultrasonic-Sensors Holder

- Motors Holders

Problem: The motor cannot be secured while the wheel is rotating, as shown in Figure 5.

Solution: A redesign is required, as shown in Figure 6.

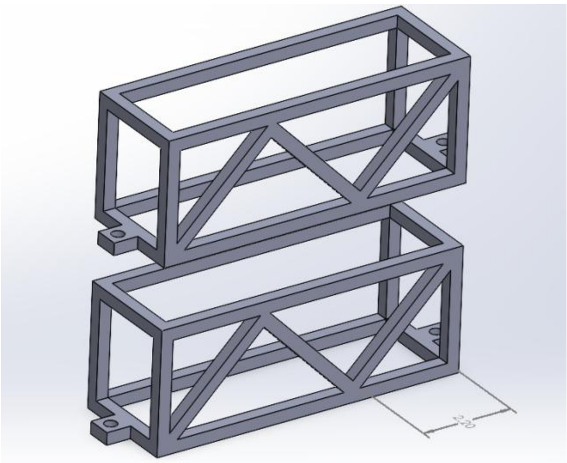

Figure 5: original design of Motors Holders

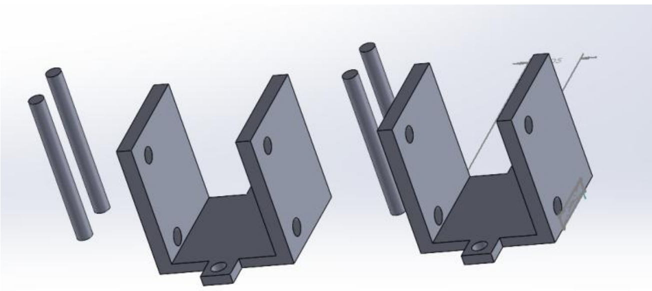

Figure 6: latest design of Motors Holders

- Battery Holder

Problem: Printing time exceeds the specified limit, as shown in Figure 7.

Solution: A redesign is required, as shown in Figure 8.

Figure 7: original design of Battery Holder

Figure 8: latest design of Battery Holder

Result

All four holders were 3D printed, assembled with their corresponding parts, and made final adjustment. The specific files have been uploaded to GitHub.